

NewsBot Intelligence System 2.0 — Executive Summary

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What is NewsBot 2.0?

NewsBot 2.0 is an end-to-end news intelligence platform that automatically reads, classifies, summarizes, and answers questions about news articles. Built on top of the mid-term NewsBot foundation, version 2.0 adds four powerful new capabilities: topic discovery, automatic summarization, multilingual support, and a conversational query interface that lets any user interact with the system using plain English.

The Problem It Solves

Modern organizations face information overload. Thousands of news articles are published every hour across global sources in multiple languages. Manually monitoring, categorizing, and extracting insights from this volume is impossible for human analysts. NewsBot 2.0 automates this entire workflow — what takes a team of analysts days takes the system seconds.

Key Capabilities

Capability	What It Means for Business	Module
Auto-Classification	Instantly categorize any article into Sport, Business, Politics, Tech, or Entertainment	M7
Topic Discovery	Find hidden themes across thousands of articles without manual labeling	M9
Smart Summarization	Get the key points of any article in 2 sentences instead of reading 500 words	M10
Semantic Search	Find relevant articles by meaning, not just keywords — like Google for your news	M10
Multilingual	Detect and translate non-English articles for global news monitoring	M11
Ask Questions	Non-technical users query the system in plain English — no coding required	M12
Sentiment Monitoring	Track positive/negative tone across categories to detect emerging issues	M6
Entity Extraction	Automatically identify people, companies, and locations mentioned in news	M8

Results at a Glance

1,490 Articles processed	92%+ Classification accuracy	30% Summary compression	5 Topics auto-discovered
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Business Applications

NewsBot 2.0 is ready for deployment in media monitoring, financial intelligence, public relations, and research organizations. The system processes articles in under one second per article and scales to thousands of daily articles. The conversational interface means non-technical staff — marketing managers, executives, analysts — can query the system without any coding knowledge. The multilingual capability enables global news coverage beyond English-language sources.

GitHub Repository

Full source code, documentation, and runnable notebooks available at:
<https://github.com/UPoojaR/PoojaUpadhyay-AI-Portfolio> in the ITAI2373-NewsBot-Midterm folder.